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ABSTRACT. It is generally believed that consumption of film and television products can induce tourism to destinations featured. We review the literature relating to this, examine it from the perspective of mass communications theory and develop a general causal model to measure the influence of screen products on visitor numbers. The model is quantified by way of a survey at two popular UK destinations. Our conclusions are that the contributions of screen products to visitor numbers are fractional, diffuse and substantial. We further find that the magnitude of the effect differs greatly between destinations and that this is related to background causes independent of screen effects.

KEYWORDS. Visitor numbers, film, television, causal modelling

INTRODUCTION

There is a general belief that the consumption of film (movie) and television products has an effect on tourism, that is, that in some way because of their consumption of movies and TV shows, people are induced to increase or otherwise change their consumption of tourism products. Many (frequently anecdotal) examples are adduced for this effect, but we do not fully understand the mechanisms that are at work in this process. This paper examines some of the assumptions common in the discussion of film and TV induced tourism and brings the mass communications literature to bear on the issues with the intention of clarifying as far as possible what these mechanisms may be. The literature mainly pursues marketing perspectives on the value of screen products in increasing visitation to the locations used, rather than examining the perspective of the audience of consumers, who choose which screen products to consume.

The paper introduces a new perspective into the literature by examining the nature of the relationships between consumption of film and TV products and consumption of tourism, as well as by providing empirical evidence about the issue. The paper examines a number of the assumptions common

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in the discussion of film and TV induced tourism in the context of the mass communications literature in the area. In particular, we question the assumption that the use of a location in a movie or television programme will immediately result in increases in visitation, and show that this is not in accordance with such theoretical approaches as are present in the literature.

We consider tourism decision-making models insofar as they address the choice of destination, in the light of the potential for movie and television influences on these decisions, using a version of the travel-buying behavior model. Taking the view that audiences choose the screen products that are most consistent with their preferences, we construct a model of the potential influence of screen products on consumers' choice of tourism destination. We assume throughout that consumers are attempting to maximize their utility through their choice of screen products and holidays.

We develop a model of the decision of the individual tourist, given his/her consumption of screen products. This model is tested using survey questionnaires that measure the contribution of films and television to the individual tourist's decision to travel to the destination location. The model permits screen products to influence the decision to travel to a screen location to some degree, if not in its entirety. This avoids the artificial classification of tourists into 'film tourist' or 'non-film tourist'. The results from the set of questionnaires are discussed and analysed and the two models, of the potential influence and of the measurement of potential influence, are evaluated in the light of the empirical evidence.

The conclusions drawn from this project are discussed. The implications both for the marketing of destinations that have the potential to be film locations and for the understanding of screen effects are considerable.

The results of the project suggest that whereas hitherto it has been assumed that tourists are either 'set-jetters' determined to visit the locations of their favorite movies or are tourists uninfluenced by their consumption of screen products, we now have evidence to show that a very high proportion of tourists are making their destination choices on the basis of some influence from screen products. Their consumption of these products may, however, have taken place long before their visit to the location becomes a reality, and may also be consequent on repeated or intensive consumption of the product which contains the images from the location.

The results show that consumption of screen products does have an effect on the rational consumer, and that this effect can be measured.

LITERATURE REVIEW

With some notable exceptions (Beeton, 2005; Tooke and Baker, 1996; Riley et al, 1998; Busby and Klug, 2001; Connell, 2005, for example) the tourism-oriented literature on film-induced tourism mainly consists of relatively unscholarly assertions of the importance of film tourism to the economy, for example the survey for Halifax Travel Insurance of just over 1,000 GB (Great Britian) adults (HBOS plc's web site "Flights, camera, action" press release, 09-08-05) which suggests that locations featured in films *and* novels can increase tourism to relevant locations by up to 30% (TNS, 2005). Many films are based on novels and short stories, as are a high proportion of TV drama features, but apart from specifically literary locations such as Haworth (the Brontës) it should be possible to separate the film effect from the book effect. Many of the anecdotal studies are mentioned in Tooke and Baker's (1996) paper which also takes four case studies from British television: *To the Manor Born, By the Sword Divided, Middlemarch* and *Heartbeat*. They also mention, for example, *Cadfael*, a historical 'detective' series for TV, which although shot in Hungary, attracted large numbers of tourists to Shrewsbury where the originating novels were set.

Riley, Baker and Van Doren (1998) gather visitation data at 12 US film locations (such as *Gettysburg, Dances with Wolves,* and *Thelma and Louise*) to assess the increase in visitation following release of the films. They estimate that the effects of the use of a location in a film can last for at least four years. From the perspective of visitation motivation they propose the concept of an 'icon', or focal point for visitation, which can be the movie's symbolic content, a single event, a favourite performer, the location's physical features or a theme. While an innovative approach, the 'icon' concept remains imprecise and seems unable to capture the full range of motivation.

Busby and Klug (2001) follow up the earlier study by discussing the challenge of measuring the 'movie-induced tourism' effect. They locate film and television tourism in the tradition of literary and other cultural tourism rather than separating out its effects. They suggest that most film tourism involves young, well-educated people. Their survey, carried out in Notting Hill, is of interest but is small scale and does not recognise that tourism within a large conurbation is likely to by influenced by many motives, whereas travel to a specific rural location can be more easily attributed to film/TV effects. Some aspects of this issue are touched on in Beeton's (2001) study of the effects of film-induced tourism on a small seaside village in Australia, where the traditional holiday market has been eclipsed by a higher-spending influx of the viewers of a TV series. Her other studies (2000, 2001a) cover the negative effects of (mainly) television induced tourism on locations which were not prepared for the influx of additional visitors or which had lost control (as they saw it) of their ability to attract the tourists they wanted, in terms of tourist type and numbers.

TV-induced effects can be seen clearly in Connell's (2005) paper, which uses a sustainability perspective to analyse the tourism generated for the Island of Mull by the children's television programme *Balamory*. By surveying the tourism businesses there in the summer of 2003, she attempts to verify the increase in visitor numbers and identify the scope and extent of impacts on businesses, including average spend, turnover, profitability and so on.

The existence of a film tourism effect is confirmed by the Travel and Tourism Analyst Report (Grihault, 2003) on International Film Tourism which suggests that a film can be used to promote lesser known regions but also emphasises that marketing campaigns need to be focused around the film's release cycle. Its main case study is *Lord of the Rings*, but it also looks at *The Beach* (Thailand) and refers to the VisitBritain survey of the impact of the first *Harry Potter* film and the National Trust's analysis of visitor numbers to their attractions, showing that visitor numbers to Alnwick Castle doubled from 2001 to 2002. Almost all attractions with *Harry Potter* connections experienced an increase in foreign visitors, with different nationals visiting different locations. An important observation from this study is that since distributors (Warner Brothers, in the case of *Harry Potter*) control not only release windows but also copyright in the promotional material, close co-operation with the distributors is necessary for effective promotion of film-induced tourism. The other case study from this report is about the *James Bond* series, and the various imaginative promotions that have developed from the films which have used locations worldwide. In a similar vein, Hanefors and Mossberg (2001) did not look directly at the effects of TV travel shows on audiences, but analysed their content to see how destination images were developed in the course of the shows and how important it was to show tourist activities on screen in order to stimulate consumer interest in new destinations. In their conclusions they point out that 'no study has yet measured the impacts….of one particular TV travel show on a destination'.

If we look for measurement of impacts, The National Trust has recently (27.09.05) released visitor figures for Basildon Park, a property featured in the new *Pride and Prejudice* film released in the previous week, showing that the usual visitor numbers of 1,300 per week had increased by 3,000 in a matter of a few days (HBOS plc's web site "Flights, camera, action" press release, 09-08-05). In a *Times* report, Dyrham Park which was used for *The Remains of the Day* in 1993 is mentioned as still attracting visitors wanting to see where filming was carried out, 12 years later, and the Giant's Causeway is mentioned as experiencing large increases in visitor numbers since its use in the TV series *Cold Feet*. These examples support the view that the film tourism effect can be considerable and long-lasting and encourage a more rigorous investigation of the evidence.

The issue of film-induced tourism has also been examined from the perspective of tourism marketing (Beeton, 2005). It is clear from her work and the examples above that the emphasis on using the media for marketing destinations is a natural extension of the belief in media effects. However, in order to understand how the viewing of a movie or TV programme translates—or does not—into visitation to a location or destination, we need to examine the literature on media effects and adopt a more analytical perspective on the issues.

There are a number of different approaches in the research into the effects of the media, and many of them are particularly concerned with researching the effects of portrayals of violence on the one hand and of advertising on the other. The first that it is useful to address is the so-called 'hypodermic needle' or Silver Bullet Model (Schramm, 1982). This perspective suggests that the media (including film and television) can as it were 'inject' ideas and messages directly into the minds of the audience and have an effect; so that, for example, using a particular location in a movie seen by a mass audience would immediately result in increased visitation to that location. This view is very popular with objectors to media portrayals of violence and with advertisers, and also with some destination managers, but is no longer taken seriously by media theorists and academic commentators because it cannot be demonstrated empirically and because it is based on assumptions about exposure to the media which cannot be sustained. We refer to this model as 'the injection model' for convenience.

There is, however, an empirical tradition in media effects research, which investigates the connection between media consumption and behavioral effects, by attempting to measure the size of audiences and the size of their reaction to the media products they consume. Within this tradition there have been waves of thought which have veered from a belief in strong effects, through a belief that the effects are in fact very weak, back to a belief that the effects of media consumption are very strong. One of the difficulties with this type of research has been that it has been very general and has assumed that all social groups have equal access to the media and equal levels of consumption, and has not taken account of market segmentation and different attitudes to media products and other types of consumption.

In contrast to the so-called 'effects model', there is a 'cultural effects' tradition, which is concerned mainly with critiques of so-called 'mass culture' and theories of hegemony and domination. Of somewhat more interest is the concept of the active audience in the Uses and Gratifications strand of research, which perceives the audience as, in the first place, selecting freely from the available media products and then consuming them according to preferences established by themselves and under the influence of peer group members and others, generating their own meanings from their 'readings' of the media. This assumption that media texts have multiple meanings should alert destination managers to the possibility that what they regard as a negative portrayal of their location might in fact generate visitation whereas a 'positive' portrayal might be ignored. John Fiske (1994), for example, would see the 'cult' or 'oppositional' readings of many media products as forces for freedom and change; however, in exploring the way in which tourism, whether 'mass' or 'individualistic', responds to the mass media, we need to distinguish between the effects of identified products (such as the *Lord of the Rings* trilogy) with huge audiences, and the incremental effects of many products over years in forming the tourism preferences and choice sets of individual consumers or segments of the tourism market.

The element of 'selectivity' on the part of the audience is clearly seen in the movie market: '…people selectively seek out information which is consonant with their pre-existing attitudes and beliefs and avoid information which is discrepant with their views' (McLeod et al., 1991), where people choose to go to the cinema, rather than consuming what happens to be available on television, or choose a DVD or video to watch from a wide range available in a library or shop. However, the ability to select from what is on offer depends on the information available to the consumer and on the consumer's ability to detect what media product is likely to be consonant with their pre-existing views. Further, the amount of attention the consumer gives to the media product will depend on many variables including the amount of time such attention is demanded, the skill with which the film or programme compels attention from the audience (or fails to) and frequency and length of exposure to the particular message being conveyed. For example, it will be interesting to see whether the single movie *Pride and Prejudice* attracts proportionately as many people to its locations as did the television series of the same name and over the same length of time.

From these brief examples it is clear that media effects research has much to offer to the debate about how film and television influence tourism consumption. We proceed to consider how this can be modelled in terms of tourists' propensity to visit film and tourism locations. The idea that the whole audience for any screen fiction will respond to it by deciding to visit its locations is clearly unlikely, so we need to define the ways in which screen consumption enters into the tourism decision-making process. One way to do this is to take the stages of consumer decision making as a guide to the opportunities for film and television to influence the selection of destinations, as is undertaken below in the following sections.

Need recognition

Tourism studies would normally consider need recognition in the light of the consumer's need for a holiday, in which case the presence of film and television as an influence would be confined to the ability of screen products to remind the audience of the need for a change of scene or activity or for relaxation. In that model, the presence of screen locations would enter the decision process at a later stage, where the location would be one of the evoked set of alternatives. We suggest, however, that in the case of location visits the need is not perceived in the general sense of 'needing a holiday' but is rather perceived as 'needing to visit X' where X represents one or more specific locations identified by the consumer as the focus of a desire to enter the screen world defined by a specific screen product or group of products. Examples might be as diverse as 'the world of Sherlock Holmes, including late Victorian London and Devon sites from *The Hound of the Baskervilles*', or '*Trainspotting* and the underworld of Scottish drug culture' or 'the Scotland of *Braveheart*, *Rob Roy* and *Kidnapped*'. A combination of screen and literature may be involved in this, when before or after viewing the screen products the consumer might also have read books which generated those products, or spin-offs generated by them. In some cases it might be possible to distinguish between screen-generated tourism and literary/book tourism, but in many cases there will be an aggregate idea of a destination compounded of the books, films, and TV programmes consumed over a lifetime. It must be a rare visitor to

Stratford-upon-Avon who has never read any part of a Shakespeare play, but an even rarer one who has not seen in the cinema, on video or on TV, one or more of *Shakespeare in Love* (1998), Olivier's *Hamlet* (1948) and *Richard III* (1955), Burton and Taylor's *Taming of the Shrew* (1967), Polanski's *Macbeth* (1971), Baz Luhrmann's *Romeo and Juliet* (1996) or even Julie Taymor's *Titus* (1999). To assume that these movies have not been part of the motivation for visiting Stratford, along with any performances the tourist has happened to see on the stage, would be as mistaken as to assume that because a visit to London has been made shortly after seeing *28 Days Later* (2002) that was the only reason for the visit.

Information search

International tourists visiting London are often disappointed to find that the fogs they have come to expect from a diet of Dickens and Conan Doyle, both read and seen, are in fact rare today. In the case of screen-induced tourism, the information search stage may be perceived as antecedent to the recognition of the need to visit, since the information has been derived either from the aggregate consumption of screen products, as may be the case with destinations such as London or Shanghai, or from one or more specific examples of destinations such as Holmfirth, location for the TV series *Last of the Summer Wine* or Basildon Park, one of the locations for *Pride and Prejudice* (2005).

As Chon (1991) explains, 'the decision maker acts upon his/her image, beliefs and perceptions of the destination rather than his/her objective reality of it' which also (Chon, 1992) includes 'the value-expressive or personality-related attributes' of the tourism product, and he clearly demonstrates the need for destination marketing to focus on both functional and symbolic attributes of the destination. Russell's (2003) examination of the cross-cultural influences of movies on consumer choice in general finds in a comparison of French and US movie-goers that US consumers have to make an effort to seek out the products of other cultures, whereas in France there is a culturally diverse cinema characterised by resistance to cultural invasion by the US. This resistance may or may not spill over to UK film products but Russell points out that 'local movies are perceived as more relevant and involving' by the French. If this effect also influences out-bound and domestic French tourism Russell's model may also be applicable to other cultural products and to the screen products (films and television programmes) of other cultures, such as the UK. As Richards (1997) suggests, films may be an important means of defining and disseminating national identity. Gammack and Hemelryk (2004) also pursue the theme of different film traditions as they compare the representation of locations in different cultures, and Jones and Smith (2005) similarly examine the development of New Zealand's national identity through the marketing campaign based on *The Lord of the Rings*. The powerful connection between literary and film tourism and the attraction of unique landscapes is further asserted by Smith (2003), and Neu (2006) illustrates the international issues that affect a destination's ability to control its exposure on film. Where the UK is concerned, Steemers (2004) quotes Collins (1986) to the effect that 'British television presents to the world a costumed image of Britain as a rigidly but harmoniously hierarchized class society', conforming to internationally current stereotypes, but the effect of this is untested.

Evaluation of alternatives

If the consumer's satisfaction is increased only by the visitation to the destination location, the only evaluation needed will be centred on the various options available for visiting that specific location. It is likely to concentrate on details of price, date, mode of transport, and quality of accommodation, instead of comparisons between destinations. The American Janeite (Jane Austen enthusiast) visiting the UK for the first time will want to include London, Bath, Chawton and

Lyme Regis in the itinerary as a minimum, with Chatsworth and Stoneleigh (where Jane Austen is known to have visited) and any other movie or TV locations as additional options. It is not our intention here to explore the commoditization of such locations or the potential conflict between education and entertainment (McKercher et al., 2004).

Purchase and post-purchase behaviour

If we accept the primacy of the location destination in the purchase decision, purchase will follow from the evaluation and selection of the preferred alternative. The level of satisfaction gained from the purchase will be determined by the consumer's experience of the location itself; that is, how well the location visit satisfies the consumer's need to enhance the screen experience by increasing the sense of reality, the feeling of 'being there', wherever 'there' happens to be. As Dicks (2003) puts it 'the image is not a substitute for the real object'; having seen a place on screen may be regarded as a prompt for people to want to visit the real rather than the imaged or imagined place.

Marketing destinations through promotions associated with movies is covered by Hudson and Richie (2006) who develop a model of the marketing inputs which informs our model of the contribution screen effects make to the tourist's destination choice. They further (2006a) show that the success of a movie and the efforts of destination marketers cannot always be directly linked to the tourism effects, thus supporting the contention which we develop below that there are many inputs to the decision to visit which may or may not be linked directly to the consumption of film or television products. Not only tourists per se but film and television producers are a target business market for destinations, as Misiura shows (2006).

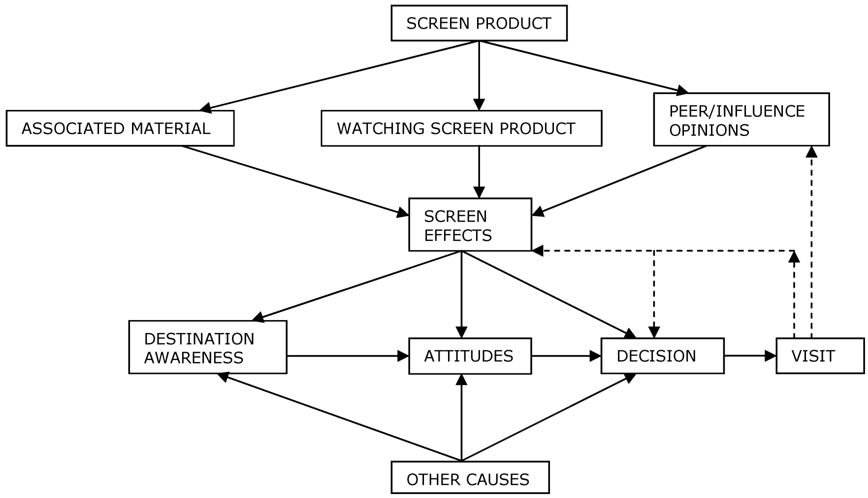
In summary, as McQuail (1975) stated (quoted in Fiske and Hartley, 1978) 'the problem in obtaining or interpreting evidence about effects in general lies partly in the inappropriateness of many formulations of mass communication, a failure to acknowledge that this is a subtle and complex process…'

THE EMPIRICAL STUDY

The model of screen induced tourism just described is incomplete. Even if the potential tourist had never seen a movie or watched television, these media might have entered into the decision to visit a particular destination through the presence of poster advertisements, brochures, and leaflets mentioning movies and TV programmes, PR in general and print media features in particular, and as a result of word-of-mouth recommendations from influencers who had already consumed the screen products. Moreover, in considering routes from a screen product (whether movie or TV) leading to a visit, one must acknowledge as part of the context that there are other causes of tourism operating alongside screen effects but quite independent of them. These additional elements are included in the model shown below in Figure 1.

The central box, Screen Effects, contains inputs from peer or reference groups as well as from advertising and press materials, so that it accommodates the possibility that tourism will be generated without the tourist actually consuming the screen product. The model also accommodates the possibility of visitation to a location which is not in any way related to the screen product but is coincidental. The visit to the location also feeds back into the reference group by word of mouth and reinforces attitudes to the location and the decision to revisit; this includes the coincidental visit, which then draws attention to the location and hence creates a revisit. An example of this effect would be where a visitor to an attraction became aware of a screen location connection while at the attraction and responded to advertising there, a brochure, an audio-visual presentation, a showing of the screen product or any combination of these, at some point leading to a re-visit. It may

FIGURE 1. Model of Screen Influences on Decision to Visit



appear that some of these effects are relatively small, but the aggregate effects on the individual in terms of raising awareness and influencing choice are as yet unmeasured. It is these effects, as well as the more direct effects of screen product consumption, that the empirical study is intended to explore.

The Theoretical Model

The first issue to be considered when constructing a survey questionnaire for use in the empirical study is that of determining exactly what the questionnaire aims to measure. Previous attempts to measure film tourism, such as Busby and Klug's (2001), are of the order of 'Have you come to (destination) to visit a place you saw in a film?' In other words, they were inviting tourists either to self-identify as film tourists or to decide that they were not. The essence of our method is to focus first on the decision of a single potential tourist. Our premise is that screen products influence this decision, not necessarily in its entirety, but in some degree.

Let V be the event that the tourist visits the destination and let S be the event of the screen product being released (including any attendant promotion and publicity). The contribution of the screen product to this tourist's visit relates to the effect of S on the probability of V, i.e.:

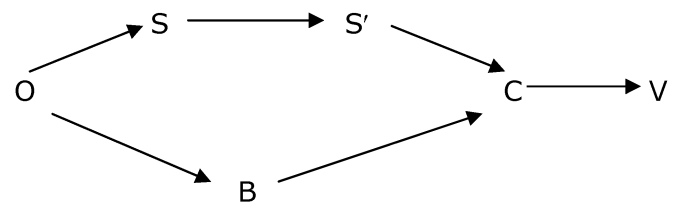
$$\Delta P \equiv P(V|S) - P(V|\sim S)$$

P(V|S) is the probability of this tourist making a visit given the event of the screen product. P(V|~S) is what the probability of a visit would have been had there not been the screen product. P(V|S)-P(V|~S) is therefore the amount that the screen product has added to the probability of this tourist making the visit. For ease of reference we call this ΔP.

The genesis of these conditional probabilities can be illustrated in the following general model of a causal chain (Figure 2).

This model of the causes of screen induced tourism is akin to a general causal chain model including background causes developed by Young, Faure and Fenn (2004). Figure 2 is intended to be understood as follows: it is to do with an account of how a visit can come to be made, in terms of what causes the visit. The point O in Figure 2 represents the origin or starting point of our explanation. A visit occurs when there is a completed path through Figure 2 from O to V. The points S′, B, and C represent events in the causal chain other than the consumption of the screen product and the visit themselves.

FIGURE 2. A General Model of a Causal Chain



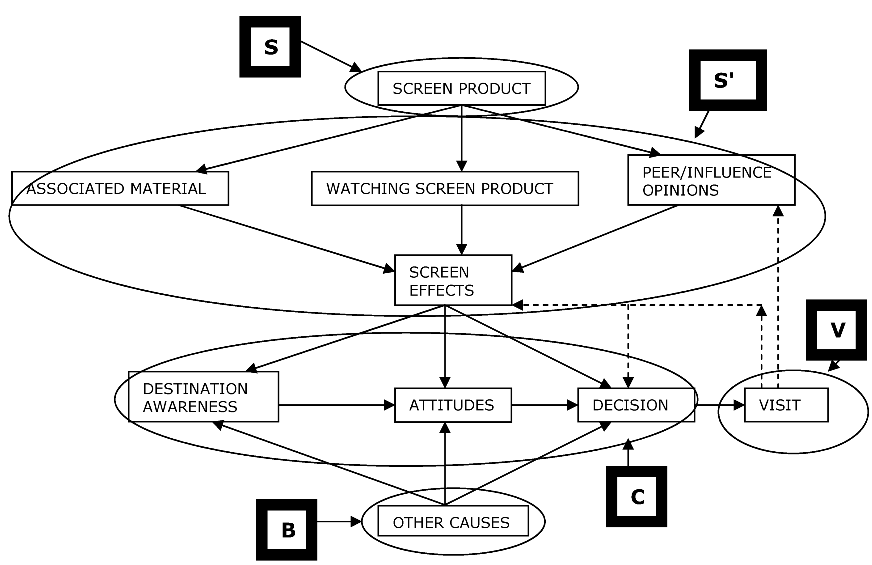
(adapted from Young, Wu and Fernandez Young (2005))

What Young and colleagues (2004) demonstrates is that whenever we speak of an event S causing an event V a causal chain with this structure or pattern is implicitly present.

This is a general model and therefore the specific model shown earlier conforms to this general structure in the way indicated in the following augmented schematic. We have used the term 'screen product' to include both movies (feature films) and television programmes, both single programmes and series. The following diagram, Figure 3, identifies the parts of Figure 1 as they form a causal chain. In other words, in Figure 3 we identify the detail of the events S′, B, and C.

The link S′ in the causal chain is in fact made up of factors associated with the screen product. B corresponds to other causes of a visit, i.e. background causes. C is made up of conversion factors, which are necessary to convert an impulse or inclination to visit into the visit itself. Thus, Figure 2 is a simplified or condensed version of Figure 3 in which we have all the detail. Figure 2 serves to make it clear that there are fundamentally two routes to a visit. V may be reached by way of a screen product, plus screen-related factors, plus conversion. Alternatively, V can be reached by way of a background cause plus conversion. In the empirical analysis which

FIGURE 3. Correspondence between Figure 1 and Figure 2



follows, this background route emerges as being of some importance.

In the survey questionnaires we measure ΔP for the respondent in the following way:

$$\Delta P = P(V|S).P(\sim V|\sim S)$$

$P(V|S)$ is, as before, the probability of visit following the screen product event. $P(\sim V|\sim S)$ is the probability that but for the screen product this tourist would not have visited. The questionnaires include questions which solicit values for $P(V|S)$ and for $P(\sim V|\sim S)$. When these are multiplied together we have a measure of the contribution of the screen product to the probability of visit. The factor $P(\sim V|\sim S)$ warrants some comment. If this factor is 0 it indicates that the tourist was certain to make the visit anyway, i.e. even had there been no screen product. In such a case, we cannot claim any contribution of the screen product to this tourist's visit. At the other extreme, if $P(\sim V|\sim S)$ is 1 it means that had this tourist not been exposed to the screen product she would definitely not have made the visit. So $P(V|S)$ and $P(\sim V|\sim S)$ each assess the effect of the screen product on the tourist in a different way and the net effect is expressed by their product, ΔP.

In extreme cases ΔP will be either 0 or 1. If ΔP is 0 the screen product has had no effect at all on the tourist's decision to visit. Where ΔP is 1 the tourist's visit is entirely motivated by the screen product. Having seen the screen product this tourist was definitely going to make the visit. In the absence of the screen product she would certainly not have made the visit.

Test results for the face to face questionnaire confirmed what one might expect: in most cases ΔP is somewhere between the extremes. The screen product has some influence; it increases the probability of visit.

The importance of measuring the effect of the screen product in this way is as follows. We are measuring the contribution of the screen product in respect of each individual tourist rather than making a binary classification of visitors into 'film' or 'non-film'. Consider the combined contribution of three hypothetical tourists, X, Y and Z. Suppose that for each of these tourists ΔP has been 0.4. Now, if ΔP is 1 we have a 'perfect' film tourist, motivated entirely by the screen product. If ΔP is 0 we have a 'perfect' non-film tourist who has not been influenced at all by the screen product. Making a binary classification would require us to classify a tourist as 'film' only if her ΔP is closer to 1 than to 0. In the cases of X, Y and Z ΔP is much closer to 0 than to 1. If we were making a binary classification, we would therefore have to classify each of X, Y and Z as 'non-film' rather than 'film'.

However, as we are measuring the screen effect we can count each of X, Y and Z as 40% of a tourist. Their combined effect is therefore 1.2 tourist visits. By measuring and accumulating the fractional contributions of the screen product, we can see that its effect on these three tourists has been to contribute the equivalent of more than one visit. If we were not looking for these fractional contributions, the effect of the screen product on these three tourists would be missed.

On a larger scale, consider a group of 100 visitors. If each of these were to have been influenced by the screen product in some degree so that her ΔP is 0.4, there would be no single tourist in the group who could properly be classified as a film tourist. However, the effect of the screen product on this group would have been equivalent to wholly motivating 40 of them. In other words, the accumulated fractional effects within this group would be equivalent to the screen product inspiring visits from 40 tourists who would not otherwise have made the visit.

In the empirical results that follow, the accumulated value of fractional effects is our summary measure of the effect of screen products on visitor numbers. For convenience of representation, we shall refer to this as 'the contribution'. In other words, when we report a contribution of x%, we mean that the aggregate of fractional influences on a group of a hundred tourists is equivalent to x out of that 100 making visits wholly motivated by screen products.

Empirical Results

A survey was carried out in which one-page questionnaires based on this perspective were used to interview tourists at the London Eye on the South Bank of the Thames in London, a notable landmark set up for the Millennium, which has been used as a location in films such as *Johnny English, Wimbledon* and television documentary series such as *Seven Ages of Britain*, and also in Oxford where *Harry Potter and the Sorcerer's Stone* (the first movie in the series) was partly located and which was the main location for the *Inspector Morse* television series. A total of 71 questionnaires were completed in Oxford over two days and 63 at the London Eye, also over two days. A copy of each of the questionnaires is included in the Appendix.

At both sites, UK residents provided the largest groups of respondents, although less than half in each case: 45% of the Oxford sample, 49% of the London Eye sample. Visitors from the rest of Europe accounted for another 25% and 22% respectively, so Europeans provided the majority of respondents. In the case of Oxford, 11% of the respondents were from Asia, against 6% of London Eye respondents. Americans were 8% and 7% at each site, Africans 4% at Oxford and 1.5% at the London Eye, Australians 5% and 1.5%. At the London Eye, 3% of respondents were Russian and 8% from the Middle East.

There was a considerable difference in the main reason for visiting the area: In the Oxford sample only 30% claimed to be visiting purely for holidays or pleasure; 44% were engaged in business, work or study, being mainly foreign students in the UK for study purposes (although not studying in Oxford). In contrast, 62% of London Eye respondents were on holiday with a further 11% visiting friends or relatives; only 24% were there for business, work or study. A substantial 17% of Oxford visitors were visiting friends or relatives. Those with other main purposes for their visit amounted to 8% of the Oxford and 3% of the London Eye samples.

Looking first at the Oxford results, in round number terms they are as follows: 20% of the visitors interviewed had either seen or heard of *Harry Potter* or *Inspector Morse*, and moreover, associated these screen products with Oxford. *Prima facie*, it would appear that this creates the potential for tourism deriving from these screen products. In the sample as a whole, the contribution of screen products was 16%. In other words, out of every 100 tourists, the contribution was 16 full tourist equivalents. This would appear to indicate that 16 out of the 20 tourists who were visiting and had made the association were visiting *because* of the association. Apparently, we have an 80% conversion rate of viewers who associate into visitors. On the one hand, a somewhat modest percentage of people make the connection from the screen product to the place but of those who do it would appear that an extremely impressive 80% go on to become tourists to that destination.

However, this interpretation has no more than surface level plausibility. On further inspection, it is entirely false. When we examine the contribution of screen products among those who make that association this turns out to be only 11%. In other words, of the 20% of people making the association, only 2% actually become tourists. This would appear to be something of a paradox. It is resolved by observing that among the people who do not make an association between either of the specific screen products and the destination, the average contribution is 17%. Among those viewers who make the association the effect in terms of contribution to visits is in fact not 80% but only 11%. This is, indeed, somewhat less than the contribution of screen products in general among those who failed to make a specific association of product and place.

The inference is that the influence of screen products is actually greater in the case of viewers who do not make a specific product to place association. This is quite contrary to the injection model of 'see, associate with the destination, make the visit'. However, it can be readily understood in terms of the diffuse causal model. Those people who have seen and associated a screen product are certainly susceptible to its influence, but, as it turns out, they are not highly susceptible. This may be because, for instance, they are not only relatively well-informed but relatively experienced and sophisticated viewers. As such, they turn out to be influenced but only in a relatively modest degree.

In contrast, there are viewers who have seen (as have all in our sample) some screen product in some way associated with Oxford, even if it is only by association of Oxford with the UK more generally. The results indicate that these viewers are more susceptible to the influence of screen products. On reflection, it may be that this is not, after all, surprising. These viewers reflect the influence of screen products in general. They may be influenced by a diffuse association of screen product with (let us say) Englishness and of Oxford with Englishness. At its most general, it may be simply that those who fail to make an association to a specific screen product are all the more susceptible to the influence of screen products in general. There is, therefore, always a distinction between being informed and being influenced.

We now turn our attention to the corresponding results at the London Eye. Approaching the London Eye results in the same way that we first approached the Oxford results, i.e. assuming an injection model, we observe first that 14% of those surveyed at the London Eye associated it with one or more of the three specific screen products. Juxtaposed with this, we have a causal contribution figure of 60%. The immediate implication of this is that the injection model is absurd. If a screen product's effect comes about only through association then we have the impossibility of explaining how screen products contributed the full tourist equivalent of 60 out of every hundred despite the fact that less than a quarter as many people made the association.

The key to resolving this contradistinction is to acknowledge a diffuse effect. At the London Eye, a substantial proportion of those surveyed expressed a very strong causal connection between their visit to the attraction and screen products in general. Many visitors said that having seen the London Eye on screen, they formed a definite intention to visit. Such expressions were, however, not attached to any named specific screen product. As many as 86% of visitors said that they had been influenced to some extent in making their visit by screen products. This general perspective on screen products is the appropriate context for understanding the 60% contribution. To complete the comparison, the 16% contribution at Oxford is in the context of 70% of tourists there having been influenced in some degree. We note that the disparity in contribution is far greater than the disparity in the percentages influenced. At the London Eye, a greater percentage of visitors had been influenced by screen products and those who had been influenced embodied a much larger degree of influence.

For ease of reference we draw together the principal results as we have described them in Table 1.

The majority of visitors to the London Eye were making their visit to London for the principal purpose of a holiday. This was the main reason for the result of 64% of visitors at the Eye, compared with 30% at Oxford. In keeping with the Oxford results, screen products made a larger contribution where the visitor was on holiday (65% v. 49%). Again, we observe that there is a substantial contribution, even among those

TABLE 1. Screen Product Effects

	OXFORD	LONDON EYE
% Associating	20	14
% Influenced	70	86
Contribution %	16	60

visitors whose primary purpose was other than a holiday.

Finally, we can compare the contribution at London Eye in the case of visitors who made a specific screen product association with those who did not. Those who associated a specific screen product (either movie or television programme) with the destination revealed a contribution of 64% by comparison with 59% among those who failed to make a specific association. Interestingly, this is the reverse of the finding at Oxford. Whereas visitors who made an association at Oxford displayed a lower contribution than those who did not, visitors to the Eye who made an association between the London Eye and its appearance on screen revealed a somewhat larger contribution than those who did not make an association. Having said this, we add the caveat that the percentage making an association was small both at the Eye and at Oxford.

Setting aside the detail, the headline result is that screen products are making a very much larger contribution to visitor numbers at the London Eye than at Oxford. In our causal model there are potentially two reasons for this. It may be that visitors to the London Eye have been more inspired by screen products to make the visit. Alternatively, it may be that visitors to Oxford were more likely to make their visit anyway, i.e. for reasons other than those susceptible to screen products. We can compare the power of these explanations by observing that the probabilities in causality and in causation are as follows: at Oxford, the probability in causality was 47%. In other words, visitors stated that having seen screen products which featured Oxford or the UK, it was 47% true to say that they were then going to visit Oxford. The corresponding figure at the London Eye was 73%, about half as large again.

Visitors to Oxford told us that it was only 28% true to say that but for screen portrayals they would not have visited Oxford. In other words, it was 72% true to say that they would have visited Oxford anyway, even if they had never seen Oxford or the UK portrayed on screen. In the case of the London Eye it was 67% true to say that but for screen product they would not have visited. Thus it is more than twice as true as not that, but for the London Eye, London or the UK featuring in screen products the visit would not have been made. The closeness of connection in this sense was almost two and a half times greater at the London Eye as at Oxford.

It would appear that we can conclude that the difference in contributions derives principally (but not exclusively) from background causes of visits. Visitors to Oxford were much more likely to have made the visit anyway, irrespective of screen portrayals. This is reinforced by examining those visitors who disclosed a total absence of influence by screen products. This was the case in respect of 30% of visitors to Oxford and 14% of visitors to the Eye. In the case of every such visitor at both locations, they expressed a definite view that they would have made the visit anyway, except for a single respondent at each location. In other words, the principal factor explaining differences in contribution rests not on screen products but on the attractiveness of the destination apart from its portrayal in screen products. The University town of Oxford is of wide international repute not only for its scholarship but for its architecture and atmosphere. It is entirely plausible that a person might form an intention to visit quite apart from any screen influence. This is rather less plausible in the case of the comparatively new attraction of the London Eye.

We now consider very briefly the issue of profiling film tourists. The pertinent results are shown in Table 2.

More precisely we consider how the contribution of screen products appears to vary according to gender and age. Given the very substantial difference in the size of contributions between Oxford and the London Eye, it is perhaps not surprising that there are also differences in profile. At Oxford, male visitors appear to have been more influenced by screen portrayals than

TABLE 2. Screen Product Contributions to Visits (ΔP) by Category

	OXFORD	LONDON EYE
Female	14	61
Male	19	59
Under 30	15	69
30–60	20	44
Over 60	17	61
UK tourists	12	53
Non-UK tourists	19	66

female visitors. The mean contribution among males is 19% compared with 14% among females. Of the three age categories in the questionnaire the middle age range displayed the largest mean contribution, 20%. Contributions in the youngest and oldest categories were respectively 15% and 17%. In other words, in Oxford the screen product contribution is greatest among male visitors and visitors in the middle age range.

At the London Eye there was no appreciable difference in terms of gender. The age group displaying the largest contribution was the youngest age group (69%) followed by the oldest age group (61%) and then the middle age range (44%). Hence, at this destination, the middle age range revealed the smallest contribution of screen products. This was still more than twice the screen product contribution at Oxford among that same age group, but is substantially less than contributions among the younger and older visitors.

Speculation as to why there are these differences in profile is beyond the purpose of the present paper, but the results are suggestive of the proposition that there is no such thing as a single consistent film tourist profile across destinations. The size of the screen product contribution depends very much on the destination and so does the distribution of the contribution over the age range and between genders.

If we compare the responses of UK with international tourists, we find results which are similar across the locations: the contribution among UK visitors to Oxford was 12% compared with 19% for overseas visitors and that from UK visitors to the London Eye was 53% compared with 66% for overseas visitors. This suggests that UK tourists are less influenced by screen products in general than are international visitors to the UK. Only three of the 63 in the London Eye sample had not seen the London Eye on television: They came from Israel, India and Lithuania.

CONCLUSIONS

In this paper we have developed a general causal model of the way in which screen products can cause tourist visits. This model recognises two types of uncertainty in the relationship: some people who consume screen products featuring a destination will not make a visit; some people who do not consume screen products will make a visit. In the latter respect, our model differs from the conventional injection way of thinking.

The principal contribution of the model to our understanding of screen tourism is that the aggregate effect is likely to be the sum of many fractional effects. A visitor may have been influenced by a screen product but less than completely and exclusively. Given that even without screen products featuring them, most (if not all) destinations could expect some tourists, the effect of presence in screen products is to contribute in some degree to the likelihood of a tourist making a visit. So, in respect of any visitor, we can expect the screen effect to be fractional—it may have contributed 20% or 60% or another fraction to the probability of that tourist's visit to the destination in question. This contrasts with the pre-existing tendency to make a binary classification of visitors into film tourists or non-film tourists.

The general causal model is particularized in terms of detailed routes leading from screen products to visits. There are many possible routes, some involving consumption of the screen product itself and some not. Some routes involve, or depend entirely on, causes other than screen products. A visitor may be influenced by material related to a screen product and by two or more screen products. The idea that this contributes to our conceptual framework is diffusion. Visits to a destination may be contributed to by many screen products and some of these may be only loosely connected to the destination. As well as considering direct effects where a major screen product features prominently a specific destination, we should look for a diffuse effect of screen products in general.

These two conceptual shifts, to fractional contributions and to diffuseness, bring with them a reorientation of perspective from screen product to destination. In addressing the connection, we must begin with the destination and ask how screen products in general have contributed in a diffuse and fractional way to visits.

From a methodological perspective, it is important to bear in mind that our theory does not assume that screen effects are necessarily fractional and diffuse, it merely acknowledges that they may be so. It does not exclude the possibility that visitors might declare themselves to be film tourists, wholly driven by association of destination with a screen specific product, or non-film tourists completely unaffected by screen products. But it does turn this possibility from an assumption into a question for empirical testing.

The survey designed around our theory was administered at two UK destinations to facilitate comparison and contrast. Oxford and the London Eye are both popular destinations, which have featured in film and television products. The results at both destinations confirmed the existence of a screen effect. The effect turns out to be fractional. The two questions in the survey allowed respondents to declare themselves to be wholly driven by screen products or completely uninfluenced. With rare exceptions (at the London Eye) they rejected this possibility and disclosed a fractional effect. Most visitors had been influenced by screen products, but in some degree only. The effects were diffuse. Few visitors recognised an association of the place they were visiting with any of the specific screen products put to them. In contrast, every respondent had seen one or more unnamed screen products connected in some way with the destination. In these qualitative ways the results from the two destinations agree and confirm that screen effects are in fact fractional and diffuse. However, quantitatively the results from Oxford and from the London Eye differed greatly. The most marked and important difference is in terms of the size of the screen contribution. This measures the screen effect in terms of the aggregate of the fractional contributions of screen products, i.e. the effect, tourist-by-tourist, aggregated over tourists as a group. The contribution shows the percentage of visits caused by screen products in full tourist equivalents. As many as 70% of visitors at Oxford acknowledged that they had been influenced in some degree by screen products. When these fractional influences are added up over a group of 100 tourists they are equivalent to 16 out of the 100 tourists being there because of screen products. In contrast, 86% of visitors to the London Eye had been influenced to some degree by screen products. The sum of the fractional influences gives a contribution figure of 60 out of 100 visitors. In other words, the fractional contributions are equivalent to 60 out of 100 visits having been caused by screen products.

These quantitative differences confirm the importance of orientating the screen-destination relationship around the destination. Our more detailed analysis using the theoretical model reveals that the reason for the difference in size of effect derived from the background causes of visits. Visitors to Oxford had other reasons for making their visit. They were generally in agreement with the proposition that but for any screen products they would have made their visit anyway. London Eye visitors were more affected by screen products because in the absence of screen associations they would have been unlikely to visit the Eye.

Having introduced in our theoretical model the possibility that screen effects might be fractional and diffuse, the empirical results confirm that this is in fact the case. The magnitude of effects is, moreover, destination-specific. These findings have implications for the evaluation of screen effects generally. It is implicit in our results that screen effects can in fact be measured, offering quantitative evidence with immediate financial implications to supplement anecdotal evidence and case studies. However, to achieve this one must look for fractional and diffuse effects in a causal setting, as we have done. Evaluation of effects must be made specific to destinations. It must take into account, as a context for the screen effect, other causes of visits to the particular destination. The screen products considered must not be limited to those where one would expect awareness of both the product and its association with the destination.

Perhaps the single most important observation for the purposes of screen effect measurement in general is the comparison, at the London Eye, of the 14% of visitors who associated the location with any of the screen products listed, with the 60% contribution of screen products to visitor numbers. It might at first appear that association is a necessary precondition for an effect. As we have shown in theory and in our empirical evidence, this is not so, because the effect is fractional and diffuse. Looking to association only would grossly underestimate the size of the screen effect. In the instance of the London Eye, the screen effect is in reality four times larger than tracing product to consumption to association to visit would suggest. There are very substantial screen effects to be found and measured, provided one goes about doing so in the right way.

REFERENCES

Beeton, S. (2000). 'It's a Wrap! What happens after the film crew leaves? An examination of community responses to film-induced tourism,' *TTRA National Conference – Lights! Camera! Action!*, Burbank, California.

Beeton, S. (2001). 'Cyclops and sirens – Demarketing as a proactive response to negative consequences of one-eyed competitive marketing', *TTRA 32nd Conference proceedings, 2001: A Tourism Odyssey*, Florida pp. 125-136.

Beeton, S. (2001). "Smiling for the camera: The influence of film audiences on a budget tourism destination." *Tourism, Culture and Communication*, 3 (1), 15-25.

Beeton, S. (2005). *Film induced tourism.* Channel View Publications.

Busby, G. and Klug, J. (2001). "Movie-induced tourism: The challenge of measurement and other issues." *Journal of Vacation Marketing*, 7 (4), 316.

Chon, K. S. (1991). 'Tourism destination image modification Process: Marketing implications,' *Tourism Management*, 12 (1), 68-72.

Chon, K. S. (1992). 'Self-image/destination-image congruity,' *Annals of Tourism Research*, 19(2), 360-363.

Collins, R. (1986). 'Wall-to-wall *Dallas*? The US-UK Trade in Television' *Screen* 66-77.

Connell, J. (2005). "'What's the story in *Balamory?*': The impacts of a children's TV programme on small tourism enterprises on the Isle of Mull, Scotland." *Journal of Sustainable Tourism*, 13 (3), 228.

Dicks, B. (2003). *Culture on display: The production of contemporary visitability.* Open University Press.

Douglas, N., Douglas, N. and Derrett, R. (eds.). (2001). *Special interest tourism: Context and cases.* Australia: John Wiley & Sons.

Fiske, J. (1989). *Reading the popular*. London and New York: Routledge.

Fiske, J. and Hartley, J. (1978). *Reading television.* London and New York: Routledge.

Gammack, J. and Hemelryk, S. (2004). 'Establishing identity: Collaborative methodologies in film and tourism,' *Proceedings of the International Tourism and Media Conference 2004,* La Trobe and Monash Universities and the University of Otago.

Grihault, N. (2003). *The travel and tourism analyst report on international film tourism,* Mintel UK.

Hanefors, M. and Mossberg, L. (2002). 'TV travel shows – A pre-taste of the destination', *Journal of Vacation Marketing* 8 (3), 235-246.

Hudson, S. and Brent Richie, J. R. (2006). 'Promoting destinations via film tourism: An empirical identification of supporting marketing initiatives,' *Journal of Travel Research* 44, May, 387.

Hudson, S. and Brent Richie, J. R. (2006). 'Film tourism and destination marketing: The case of Captain Corelli's Mandolin,' *Journal of Vacation Marketing*, 12 (3), 256.

Jones, D. and Smith, K. (2005). 'Middle-earth meets New Zealand: Authenticity and location in the making of *The Lord of the Rings,*' *Journal of Management Studies*, 42 (5), 923.

McKercher, B., du Cros, H., Ho, P. (2004). 'Attributes of popular cultural tourism attractions,' *Annals of Tourism Research* 31(2), 393:407.

McLeod, J., Kosicki, G. M. and Zhongdang P. (1991). 'On understanding and misunderstanding media

effects' in *Mass Media and Society* Curran, J. and Gurevitch, M. (eds.). London: Edward Arnold.
McQuail, D. (1975). *Communication* London: Longmans.
Misiura, S. (2006). *Heritage marketing.* Butterworth-Heinemann.
Neu, T. (2006). 'Bollywood is coming! Copyright and film industry issues regarding international film co-productions involving India,' *Cornell Law School LL.M. Papers Series,* 14.
Richards, J. (1997). *Films and British national identity: From Dickens to dad's army,* Studies in Popular Culture Series, Manchester University Press.
Riley, R., Baker, D. and Van Doren, C. S. (1998). "Movie induced tourism," *Annals of Tourism Research*, 25 (4), 919-935.
Russell, C. A. (2003). "Now showing: Global movies crossing cultural lines. Is resistance futile?" *Proceedings of the Cross Cultural Research Conference*, Seren Askegaard ed.
Schramm, W. (1982). *Men, women, messages and media: Understanding human communication.* 2nd ed. Harper.
Tooke, N. and Baker, M. (1996). "Seeing is believing: The effect of film on visitor numbers to screened locations." *Tourism Management*, 17 (2), 87-94.
Young, R., Faure, M. and Fenn, P. (2004). 'Causality and causation in tort law,' *International Review of Law and Economics,* 24 (4), 507.
Young, R., Wu, M. J. and Fernandez Young, A. (2006). 'Causing tourism,' *Proceedings of the 4th De Haan Tourism Management Conference* Nottingham University Business School Discussion Papers.

APPENDIX: SURVEY QUESTIONNAIRES

We are researching how movies and television influence tourists' choice of destinations. We would appreciate a few moments of your time to answer this short survey.

Please tell us where your home is:
Gender F ☐ M ☐ Age group: Under 30 ☐ 30-60 ☐ Over 60 ☐

What is the main reason for your visit to this area? Please mark ONE answer only.
Holiday/pleasure ☐ Visiting family or friends ☐ Business/work/study ☐ Other ☐

Have you seen any of the following portrayed in a movie or on television? Please mark as many of the boxes as apply.

	I saw this....	
	in a **movie**	on **television**
Oxford		
Another part of the UK		

If the respondent has not seen any relevant screen product, terminate the interview now.

We are interested in any connection between what you have seen on screen and your visit to Oxford.

Please indicate to what extent you agree with each statement.

Once I had seen Oxford on screen, I had to come.

Disagree Agree

0	1	2	3	4	5	6	7	8	9	10

What I saw on screen made no difference at all. I was going to visit Oxford anyway.

Disagree Agree

0	1	2	3	4	5	6	7	8	9	10

We'd like to ask you about two specific products:

Product	Have you seen?	Have you heard of?	Do you associate with Oxford?
First Harry Potter Movie			
Inspector Morse detective series			

Is there anywhere you would like to visit because you saw it on screen?

Place________Movie/TVShow________

We are researching how movies and television influence tourists' choice of destinations. We would appreciate a few moments of your time to answer this short survey.

Please tell us where your home is: ________________________________

Gender F ☐ M ☐ Age group: Under 30 ☐ 30-60 ☐ Over 60 ☐

What is the main reason for your visit to this area? Please mark ONE answer only.

Holiday/pleasure ☐ Visiting family or friends ☐ Business/work/study ☐ Other ☐

Have you seen any of the following portrayed in a movie or on television? Please mark as many of the boxes as apply.

	I saw this....	
	in a **movie**	on **television**
The London Eye		
Any part of London		
Another part of the UK		

If the respondent has not seen any relevant screen product, terminate the interview now.

We are interested in any connection between what you have seen on screen and your visit to Oxford.

Please indicate to what extent you agree with each statement.

Once I had seen the London Eye on screen, I had to come.

Disagree Agree

0	1	2	3	4	5	6	7	8	9	10

What I saw on screen made no difference at all. I was going to visit Oxford anyway.

Disagree Agree

0	1	2	3	4	5	6	7	8	9	10

We'd like to ask you about three specific products:

Product	Have you seen?	Have you heard of?	Do you associate with the Eye?
Johnny English (Movie)			
Wimbledon (Movie)			
Seven Ages of Britain (TV doc)			

Is there anywhere you would like to visit because you saw it on screen?

Place__________Movie/TVShow________